

Priprema za LV iz Dizajna kompajlera - Regularni
izrazi

Za izradu vježbi koristimo interaktivni regex simulator. Ako na LV računarima imate internet otvorite sljedeći link: <https://regex101.com>. Za pokretanje regex101 lokalno (u slučaju da internet nije dostupan) pokrenuti fet-base kontejner, navigirati u direktorij sa pripremom za vježbe i pokrenuti skriptu `run_local.sh` i navigirati na <https://localhost:8000>.

Odabrati PCRE (PHP) sintaksu regularnih izraza!

Regex cheat sheet

Regular Expression Basics	Regular Expression Character Classes	Regular Expression Flags
.	[ab-d]	i
a	[^ab-d]	m
ab	[b]	s
a b	\d	x
a*	\D	J
\	\s	U
	\S	(?Lmsux) Set flags within regex
	\w	
	\W	
Regular Expression Quantifiers	Regular Expression Assertions	Regular Expression Special Characters
*	^	\n
+	\A	\r
?	\$	\t
{2}	\Z	\0
{2, 5}	\b	\YYY
{2,}	\B	\xYY
	\G	\x{YY}
Default is greedy. Append ? for reluctant.	(?=...)	\cY
	(?!...)	
	(?<=...)	
	(?<!....)	
	(?)	
Regular Expression Groups	Regular Expression Escapes	Regular Expression Posix Classes
(...)	\Q...E	[alnum:]
(?P<Y>...)		[alpha:]
(?:...)		[ascii:]
(?>...)		[blank:]
(?[...)		[cntrl:]
\Y		[digit:]
(?P=Y)		[graph:]
(?R)		[lower:]
(?Y)		[print:]
(?&Y)		[punct:]
\g{Y}		[space:]
\g<Y>		[upper:]
(?#...)		[word:]
		[xdigit:]

Figure 1: Cheat sheet

Zadatak 1.

Napisati regularni izraz koji matchira heksadecimalne brojeve. Primjeri ulaza:

```
0x0
0x1234
0xabcdef
0xca11ab1e
```

```
208
0xtar
```

Zadatak 2.

Napisati regularni izraz koji matchira validne identifikatore prema C/CPP programskom jeziku. Primjeri ulaza:

```
mojInteger
index21
foo_
_bar
_bar42
```

```
10ducks
page-break
```

Zadatak 3.

Napisati regularni izraz koji matchira decimalne brojeve. Primjeri ulaza:

```
0
1
12
+10
-200
401.5
+12.615
-4.5e10
+12.515E-8
```

```
25+1.2
100.3.5
-1.-5e10
```

Zadatak 4.

Napisati regularni izraz koji matchira brojeve telefona. Primjeri ulaza:

```
061123456
```

061-123-456
066 123 456
+38762123456
+387-66-423-156
+387 66 423 156

Zadatak 5.

Dat je dio html koda sa neke stranice. Napisati regularni izraz koji će pronaći http linkove unutar html koda. Dovoljno je naći linkove koji se nalaze pod href tagom.

```

<div class="portal" role="navigation" id='p-navigation'>
<h3>Navigation</h3>
<div class="body">
<ul>
<li id="n-mainpage-description">
<a href="/wiki/Main_Page" title="Visit the main page [z]" accesskey="z">Main page</a></li>
<li id="n-contents">
<a href="/wiki/Portal:Contents" title="Guides to browsing Wikipedia">Contents</a></li>
<li id="n-featuredcontent">
<a href="/wiki/Portal:Featured_content" title="Featured content the best of Wikipedia">
Featured content</a></li>
<li id="n-currentevents">
<a href="/wiki/Portal:Current_events" title="Find background information on current events">
Current events</a></li>
<li id="n-randompage">
<a href="/wiki/Special:Random" title="Load a random article [x]" accesskey="x">Random article
</a></li>
<li id="n-sitesupport">
<a href="//donate.wikimedia.org/wiki/Special:Fundraiser?uselang=en" title="Support us">
Donate to Wikipedia</a></li>
</ul>
</div>
</div>

```

Grep

grep je CLI alat za pretragu saržaja fileova na file sistemu. Spremiti gornji html kao file i pokrenuti komandu:

```
grep -P 'href=".*"' index.html
```

ili

```
grep -P 'href=".*"' -R .
```

WHENEVER I LEARN A
NEW SKILL I CONCOCT
ELABORATE FANTASY
SCENARIOS WHERE IT
LETS ME SAVE THE DAY.

OH NO! THE KILLER
MUST HAVE FOLLOWED
HER ON VACATION!



BUT TO FIND THEM WE'D HAVE TO SEARCH
THROUGH 200 MB OF EMAILS LOOKING FOR
SOMETHING FORMATTED LIKE AN ADDRESS!

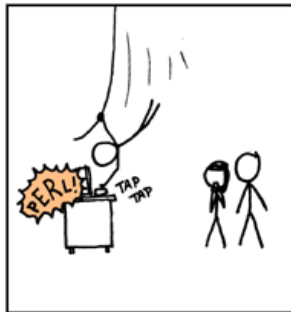
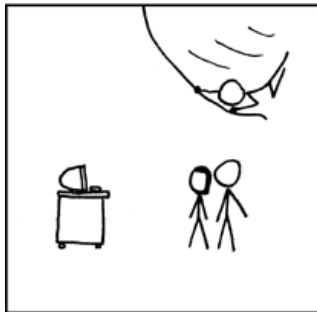


IT'S HOPELESS!

EVERYBODY STAND BACK.



I KNOW REGULAR
EXPRESSIONS.



Source xkcd 208